**Start Date: 18.12.2025**
***Due Date: 05.01.2026, 23:55***

# CENG467
# PROJECT 2

## Overview for Data-to-Text Generation Task:

**Data-to-Text Generation** is a fundamental task in Natural Language Generation (NLG) that focuses on producing fluent, natural language descriptions from structured data. This structured input can take various forms, such as tables, Knowledge Graphs, or database records. The primary goal is to translate this non-linguistic data into coherent text that accurately reflects the information provided. This capability is crucial for a wide range of downstream applications, including open-domain question answering, dialogue systems, and digital assistants.

For this assignment, we will focus on the **ToTTo Dataset** (Table-to-Text), a benchmark dataset comprised of Wikipedia tables paired with human-written descriptions. Unlike generic table-to-text tasks, ToTTo presents a controlled generation challenge: each input consists of a full table with specific **highlighted cells**. Your models are required to generate text that faithfully describes *only* these highlighted cells, avoiding the "hallucination" of extra information not present in the source or the omission of key selected details. The original repository of ToTTo can be found through the following link: https://github.com/google-research-datasets/ToTTo. The version of ToTTo dataset we will use can be accessed from here: https://huggingface.co/datasets/GEM/totto.

To understand how advanced models tackle this, you can refer to the paper *"Plan-then-Generate: Controlled Data-to-Text Generation via Planning"*. You can access the Github repository of Plan-then-Generate using the following link: GitHub - yxuansu/PlanGen: [EMNLP'21] Plan-then-Generate: Controlled Data-to-Text Generation via Planning. This repository also provides an example Jupyter Notebook file that show evaluation on the ToTTo dataset: PlanGen/generator/Controlled Generation.ipynb at main · yxuansu/PlanGen · GitHub.

While the codes for this project do not require you to use Plan-then-Generate approach, this repository may help you understand this task better (especially which metrics are used for this task). Unlike the example paper, we will use an LLM to perform evaluations for this task. Our LLM of choice in this project is **"Qwen/Qwen3-4B".** (https://huggingface.co/Qwen/Qwen3-4B)

We will evaluate our inference results using the following metrics: BLEU and BLEURT. There is also the PARENT metric, but it is deprecated. So, we will not use it.

# Project Questions:

All question answers MUST BE HANDWRITTEN ON AN A4 PAPER. We will not accept any homeworks written digitally on Word or any similar program.

Recommended Python Environment Requirements:

*Python==3.12.11*
*PyTorch (you may use the same torch version from the Project 1)*
*pip install datasets==3.6.0*
*pip install transformers==4.57.0*
*pip install bleurt-pytorch*
*pip install sacrebleu*
*pip install protobuf*

## Q1)

(20 pts) Write a code that uses the "Qwen/Qwen3-4B" model to take inputs from the ToTTo dataset and output candidate texts that describe a highlighted cell of the tables inside the dataset. The highlighted cell position can be found inside the "[highlighted_cells]" column of the dataset. The complete tables are found in the "[table]" columns of the dataset. Compare the candidate texts with the ground truth sentences in ToTTo (found under ['target'] column). Use BLEU and BLEURT metrics to measure the success of the candidate texts generated from the tables.

You cannot directly pass the tables to the model as they are provided in the dataset in json format. For Q1, you need to transform them into a special markdown structure and then pass them to the model.

Here is the initial part of an example table from ToTTo (its highlighted cell is [[13, 2]]):

```
'table': [[{'column_span': 1, 'is_header': True, 'row_span': 1, 'value': '#'},
   {'column_span': 1, 'is_header': True, 'row_span': 1, 'value': 'Run'},
   {'column_span': 1, 'is_header': True, 'row_span': 1, 'value': 'Title'},
   {'column_span': 1, 'is_header': True, 'row_span': 1, 'value': 'Chapters'},
   {'column_span': 1, 'is_header': True, 'row_span': 1, 'value': 'Author'},
   {'column_span': 1, 'is_header': True, 'row_span': 1, 'value': 'Director'},
   {'column_span': 1,
    'is_header': True,
    'row_span': 1,
    'value': 'Ibope Rating'}],
  [{'column_span': 1, 'is_header': False, 'row_span': 1, 'value': '59'},
   {'column_span': 1,
    'is_header': False,
    'row_span': 1,
    'value': 'June 5, 2000– February 2, 2001'},
   {'column_span': 1,
    'is_header': False,
    'row_span': 1,
    'value': 'Laços de Família'},
   {'column_span': 1, 'is_header': False, 'row_span': 1, 'value': '209'},
   {'column_span': 1,
    'is_header': False,
    'row_span': 1,
    'value': 'Manoel Carlos'},
   {'column_span': 1,
    'is_header': False,
    'row_span': 1,
    'value': 'Ricardo Waddington'},
```

```
  {'column_span': 1, 'is_header': False, 'row_span': 1, 'value': '44.9'}],
 [{'column_span': 1, 'is_header': False, 'row_span': 1, 'value': '60'},
  {'column_span': 1,
   'is_header': False,
   'row_span': 1,
   'value': 'February 5, 2001– September 28, 2001'},
  {'column_span': 1,
   'is_header': False,
   'row_span': 1,
   'value': 'Porto dos Milagres'},
  {'column_span': 1, 'is_header': False, 'row_span': 1, 'value': '203'},
  {'column_span': 1,
   'is_header': False,
   'row_span': 1,
   'value': 'Aguinaldo Silva Ricardo Linhares'},
        ...
        ...
```

Here is what it should look like after being transformed into the markdown format (with special <highlight> markers to inform the model):

```
| # | Run | Title | Chapters | Author | Director | Ibope Rating |
| --- | --- | --- | --- | --- | --- | --- |
| 59 | June 5, 2000– February 2, 2001 | Laços de Família | 209 | Manoel Carlos | Ricardo Waddington | 44.9 |
| 60 | February 5, 2001– September 28, 2001 | Porto dos Milagres | 203 | Aguinaldo Silva Ricardo Linhares | Marcos Paulo Simões | 44.6 |
| 61 | October 1, 2001– June 14, 2002 | O Clone | 221 | Glória Perez | Jayme Monjardim | 47.0 |
| 62 | June 17, 2002– February 14, 2003 | Esperança | 209 | Benedito Ruy Barbosa | Luiz Fernando | 37.7 |
| 63 | February 17, 2003– October 10, 2003 | Mulheres Apaixonadas | 203 | Manoel Carlos | Ricardo Waddington | 46.6 |
| 64 | October 13, 2003– June 25, 2004 | Celebridade | 221 | Gilberto Braga | Dennis Carvalho | 46.0 |
| 65 | June 28, 2004– March 11, 2005 | Senhora do Destino | 221 | Aguinaldo Silva | Wolf Maya | 50.4 |
| 66 | March 14, 2005– November 4, 2005 | América | 203 | Glória Perez | Jayme Monjardim Marcos Schechtman | 49.4 |
| 67 | November 7, 2005– July 7, 2006 | Belíssima | 209 | Silvio de Abreu | Denise Saraceni | 48.5 |
| 68 | July 10, 2006– March 2, 2007 | Páginas da Vida | 203 | Manoel Carlos | Jayme Monjardim | 46.8 |
| 69 | March 5, 2007– September 28, 2007 | Paraíso Tropical | 179 | Gilberto Braga Ricardo Linhares | Dennis Carvalho | 42.8 |
| 70 | October 1, 2007– May 31, 2008 | Duas Caras | 210 | Aguinaldo Silva | Wolf Maya | 41.1 |
| 71 | June 2, 2008– January 16, 2009 | <highlight>A Favorita</highlight> | 197 | João Emanuel Carneiro | Ricardo Waddington | 39.5 |
| 72 | January 19, 2009– September 11, 2009 | Caminho das Índias | 203 | Glória Perez | Marcos Schechtman | 38.8 |
| 73 | September 14, 2009– May 14, 2010 | Viver a Vida | 209 | Manoel Carlos | Jayme Monjardim | 35.6 |
```

Additionally, you need to pass to the model other auxiliary information provided by ToTTo. These are table_page_title, table_section_title, and table_section_text. The latter two may not be available for each example in the dataset. If they are not available, they do not need to be present in the prompt to the LLM. Here is the relevant auxiliary information for the example above:

```
table_page_title: List of 8/9 PM telenovelas of Rede Globo
table_section_title: 2000s
table_section_text:  -
```

The target (ground truth) sentence of this example is "`A Favorita is the telenovela aired in the 9 pm timeslot.`".

For the evaluation operations, use the validation set of ToTTo since the test set hides the ground truth final_sentence to prevent cheating in the leaderboards. This task has a public leaderboard you can view in its GitHub repo. It is enough for you to directly use the validation set for this project since it is easily accessible.

Note that validation set has 7700 examples. To speed up our runtimes, we will use only the first 300 examples/rows in the validation set for this project. You can use Google Colab if your local GPU is not enough to utilize Qwen/Qwen3-4B.

Below, I will provide the prompt you will utilize for this task. It has been inspired by the "Plan-then-Generate" paper's findings by instructing the model to focus strictly on the **highlighted cells** to ensure faithful generation. Prompt will have 2 examples (few-shot) to help model

understand the task better. Bottom part of the prompt will be incomplete since you will need to concatenate each example in the validation set after the few-shot examples.

Make the generation max token size 50. We expect only a single output sentence per example. Also, there may be situations where generated outputs may not fit to the format we are expecting since the model size is small. You can make minor regex manipulations to only extract the desired single sentence outputs if necessary.

Here is the prompt:

```
Your task is to generate a fluent sentence that describes the highlighted information in the provided table.

**Instructions:**
1. Read the **Input Table** and the **Auxiliary Context** (Page Title, Section Title, Section Text).
2. Focus **only** on the cells wrapped in `<highlight>` and `</highlight>` tags.
3. Write a coherent, factual sentence that incorporates the information from these highlighted cells.
4. You may use the Page Title or Section Title to provide necessary context (e.g., to identify the subject), but do not
hallucinate details or include information from non-highlighted cells.
5. ONLY give a single sentence answer after the prompt.

---

### Example 1

**Input Table:**
| # | Run | Title | Chapters | Author | Director | Ibope Rating |
| --- | --- | --- | --- | --- | --- | --- |
| 59 | June 5, 2000– February 2, 2001 | Laços de Família | 209 | Manoel Carlos | Ricardo Waddington | 44.9 |
| 60 | February 5, 2001– September 28, 2001 | Porto dos Milagres | 203 | Aguinaldo Silva Ricardo Linhares | Marcos Paulo Simões
| 44.6 |
| 61 | October 1, 2001– June 14, 2002 | O Clone | 221 | Glória Perez | Jayme Monjardim | 47.0 |
| 62 | June 17, 2002– February 14, 2003 | Esperança | 209 | Benedito Ruy Barbosa | Luiz Fernando | 37.7 |
| 63 | February 17, 2003– October 10, 2003 | Mulheres Apaixonadas | 203 | Manoel Carlos | Ricardo Waddington | 46.6 |
| 64 | October 13, 2003– June 25, 2004 | Celebridade | 221 | Gilberto Braga | Dennis Carvalho | 46.0 |
| 65 | June 28, 2004– March 11, 2005 | Senhora do Destino | 221 | Aguinaldo Silva | Wolf Maya | 50.4 |
| 66 | March 14, 2005– November 4, 2005 | América | 203 | Glória Perez | Jayme Monjardim Marcos Schechtman | 49.4 |
| 67 | November 7, 2005– July 7, 2006 | Belíssima | 209 | Sílvio de Abreu | Denise Saraceni | 48.5 |
| 68 | July 10, 2006– March 2, 2007 | Páginas da Vida | 203 | Manoel Carlos | Jayme Monjardim | 46.8 |
| 69 | March 5, 2007– September 28, 2007 | Paraíso Tropical | 179 | Gilberto Braga Ricardo Linhares | Dennis Carvalho | 42.8 |
| 70 | October 1, 2007– May 31, 2008 | Duas Caras | 210 | Aguinaldo Silva | Wolf Maya | 41.1 |
| 71 | June 2, 2008– January 16, 2009 | <highlight>A Favorita</highlight> | 197 | João Emanuel Carneiro | Ricardo Waddington |
39.5 |
| 72 | January 19, 2009– September 11, 2009 | Caminho das Índias | 203 | Glória Perez | Marcos Schechtman | 38.8 |
| 73 | September 14, 2009– May 14, 2010 | Viver a Vida | 209 | Manoel Carlos | Jayme Monjardim | 35.6 |

**Auxiliary Context:**
* Page Title: List of 8/9 PM telenovelas of Rede Globo
* Section Title: 2000s
* Section Text: -

**Target Sentence:**
A Favorita is the telenovela aired in the 9 pm timeslot.

---

### Example 2

**Input Table:**
| State (class) | Vacator | Reason for change | Successor | Date of successor's formal installation |
| --- | --- | --- | --- | --- |
| New Jersey (1) | <highlight>William Warren Barbour (R)</highlight> | <highlight>Died November 22, 1943. Successor was appointed
until an election.</highlight> | <highlight>Arthur Walsh (D)</highlight> | <highlight>November 26, 1943</highlight> |
| Indiana (3) | Frederick Van Nuys (D) | Died January 25, 1944. Successor was appointed until an election. | Samuel D. Jackson
(D) | January 28, 1944 |
| Massachusetts (2) | Henry Cabot Lodge, Jr. (R) | Resigned February 3, 1944, to go on active duty in the US Army. Success was
appointed until a special election. | Sinclair Weeks (R) | February 8, 1944 |
| Oregon (2) | Charles L. McNary (R) | Died February 25, 1944. Successor was appointed and subsequently won special election |
Guy Cordon (R) | March 4, 1944 |
| Washington (3) | Homer Bone (D) | Resigned November 13, 1944, to become Judge of the U.S. Court of Appeals for the 9th Circuit.
| Warren Magnuson (D) | December 14, 1944 |
| Indiana (3) | Samuel D. Jackson (D) | Appointee was not a candidate to finish the term. Successor was elected. | William E.
Jenner (R) | November 7, 1944 |
| South Carolina (3) | Ellison Durant Smith (D) | Died November 17, 1944. | Wilton E. Hall (D) | November 20, 1944 |
| New Jersey (1) | <highlight>Arthur Walsh (D)</highlight> | Appointee was not a candidate to finish the term. Successor was
elected. | <highlight>H. Alexander Smith (R)</highlight> | <highlight>December 7, 1944</highlight> |
| Massachusetts (2) | Sinclair Weeks (R) | Appointee was not a candidate to finish the term. Successor was elected. | Leverett
Saltonstall (R) | January 4, 1945 |
```

```
**Auxiliary Context:**
* Page Title: 78th United States Congress
* Section Title: Senate
* Section Text: -

**Target Sentence:**
William Warren Barbour (R) served as Senate until November 22, 1943, then by Arthur Walsh (D) from November 26, 1943 and was
succeeded by H. Alexander Smith (R), from December 7, 1944.

---

### Current Task

**Input Table:**
```

Write your evaluation code inside a Python file named "CENG467_Project2_Q1.py". The output sentences and the ground truth sentences should be recorded inside a Json file named Q1_outputs.json. This rule applies to the remaining code-related question as well.

*(You may want to be careful about the size of the prompts. The prompts may be too long for the input size limitations of QWEN. You can truncate your input token size to a value like 4096.)*

# Q2)

a) (5 pts) Observe the metrics you have found from Q1 and write them down in your report.

b) (10 pts) Comment on how QWEN performed compared to the previous approaches' results shown in the task leaderboard (https://github.com/google-research-datasets/ToTTo#leaderboard). How did QWEN perform compared to them? Why?

c) (5 pts) Find at least two examples which are very similar (even almost identical) to the ground truth target. Try to make guesses about why QWEN worked for them so well compared to other examples. Make sure you write down all details of your examples in the report.

d) (10 pts) Find at least two examples where QWEN failed to generate an acceptable output. For example, hallucinated facts in the output could be one reason because QWEN fails. Comment on why the chosen outputs are not good enough. Why could have QWEN failed for these examples?

e) (5 pts) What is the difference between BLEU and BLEURT score? You can research the difference online. Then, explain which score is better for this task. Do you think one of the scores may fail to reflect the successful outputs from QWEN model?

f) (5 pts) Suggest potential improvements to the prompt we gave to QWEN. Explain why your idea might be good for QWEN.

g) (20 pts) Re-run your code with your own prompt using a new file named "CENG467_Project2_Q2_g.py". Write down your results and compare them to the previous results. Comment on the effect of your custom prompt on the results. Why?

## Q3)

a) (5 pts) You will now change how you transform the table data to markdown format. In addition to highlighting elements inside the cells, you will also add special **<important>** markers at the start of the relevant rows that have highlighted cells. You should also adapt the examples in the prompt to reflect these changes. Also, change the rules 2 and 3 as follows:

```
2. Focus **only** on the rows marked with `<important>` tags. Specifically,
focus on cells wrapped in `<highlight>` and `</highlight>` tags.
3. Write a coherent, factual sentence that incorporates the information from
these important rows with the highlighted cells.
```

Re-run this version of the code in a file named "CENG467_Project2_Q3.py".

b) (10 pts) Write down the output metrics of the previous step. Comment on the results. What are the changes compared to before? Why?

c) (5 pts) Observe the output sentences and select one example with significant hallucinations. Explain why this result may be caused by the changes in this version of the code.

# Submission Rules:

- If you are using Colab, we recommend that you move your entire code to Google Drive and mount the corresponding Drive disk to run the code. Do not try to copy and paste the entire code on a Colab project as it may cause issues.
- All questions (that require mathematical operations or writing down answers/comments) MUST BE HANDWRITTEN ON AN A4 PAPER. We will not accept any project reports written digitally on Word, etc.
- You are expected to submit a project report named "CENG467_project2_report_<groupnumber>.pdf". The contents of this report should be handwritten and scanned using an app like CamScanner. Make sure that the top-right corner of the first page has the group no, names and numbers of your group members.
- You should also submit the entire codebase (after you have filled the STUB sections) in a zip file named "CENG467_project2_code_<groupnumber>.zip". Make sure that your

group no, your names and numbers are written (in comments) on the top of each code file you performed modifications on.

- In Teams, it is enough for only a single group member to upload these 2 files. (For example, "*CENG467_project2_report_G07.pdf*" and "*CENG467_project2_code_G07.zip*")
- Collaboration between different groups is not allowed.